

Homework B2 | ECON 0100 | Fall 2026

Due: Sunday, Sep. 20

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely, and show your work so you can later understand your thought process. You are welcomed and encouraged to work in groups as long as your work is your own. Submit your answers on Gradescope when you're finished: each answer there is a selection.

Butterbeer

The supply curve for butter beer can be represented by the following equation:

$$P = 20 + \frac{1}{2}Q$$

Prices are in galleons and quantity is in bottles.

Q1 | Quantity Supplied

Use a graph to plot this supply curve, including the quantity supplied at both 40 galleons and 50 galleons.

- a) Quantity supplied at 40 galleons: _____
- b) Quantity supplied at 50 galleons: _____
- c) How much did quantity supplied change as the price dropped from 50 to 40? _____
- d) What is marginal cost at a quantity of 60? _____

Q2 | Producer Surplus

Then find and label the producer surplus at these prices.

a) PS at 40 galleons: _____

b) PS at 50 galleons: _____

c) How much did producer surplus change as the price dropped from 50 to 40? _____